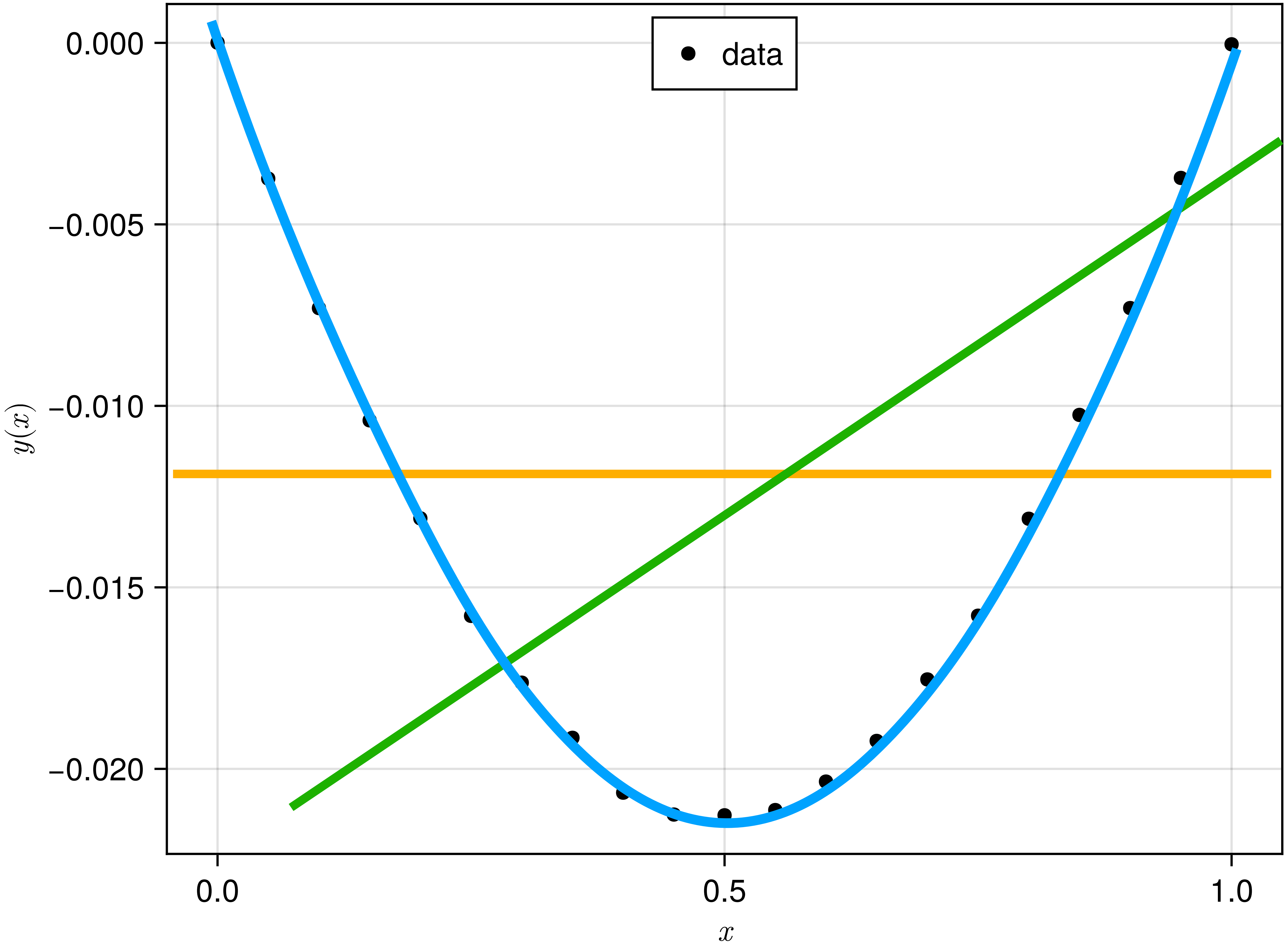
Suppose you are given some data



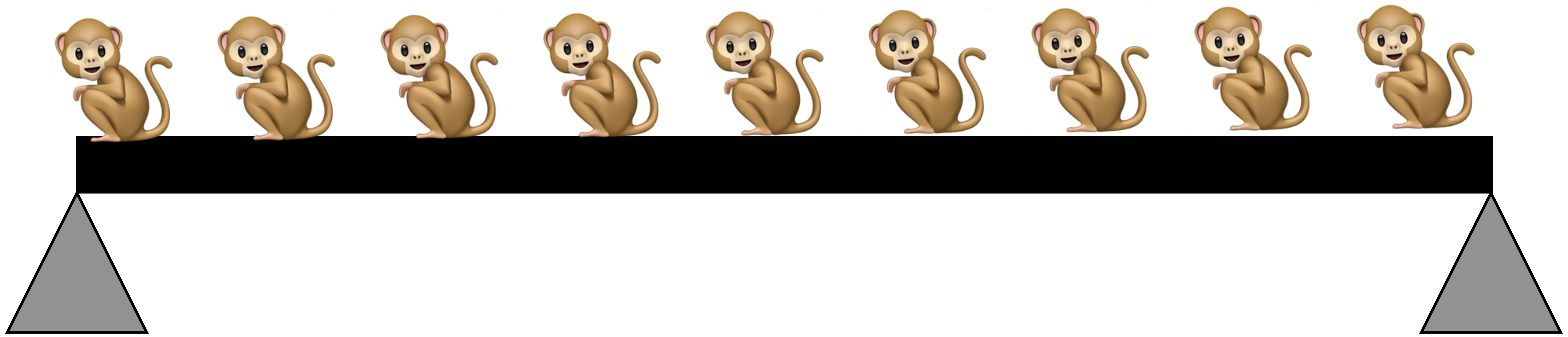
$f_{model}(\vec{a}, x) = a_0$

$f_{model}(\vec{a}, x) = a_0 + a_1 x$

$f_{model}(\vec{a}, x) = a_0 + a_1 x + a_2 x^2 + \ldots.$

You need to guess $f_{model}(\vec{a}, x)$ looks like. Once you guess, you can evaluate the value(s) of $f_{model}(\vec{a}, x)$

If I were to tell you that the data comes from a beam that is deflected by some monkeys sitting on it!

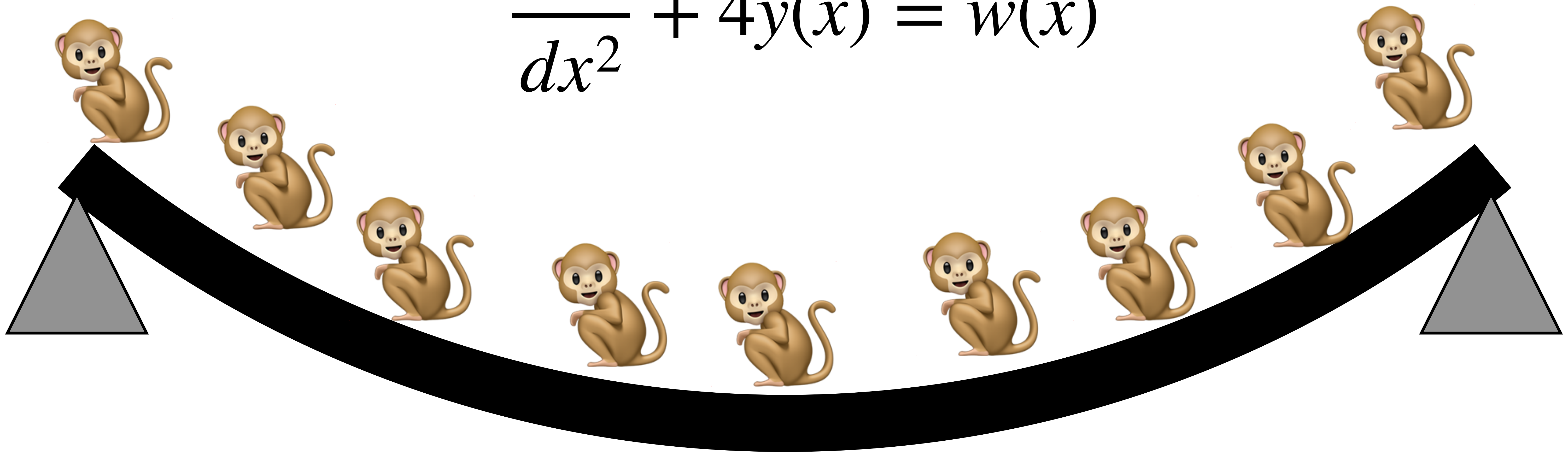


If I were to tell you that the data comes from a beam that is deflected by some monkeys sitting on it!

$y(x)$ is the deflection of the beam at location x

$$\frac{d^2y}{dx^2} + 4y(x) = w(x)$$

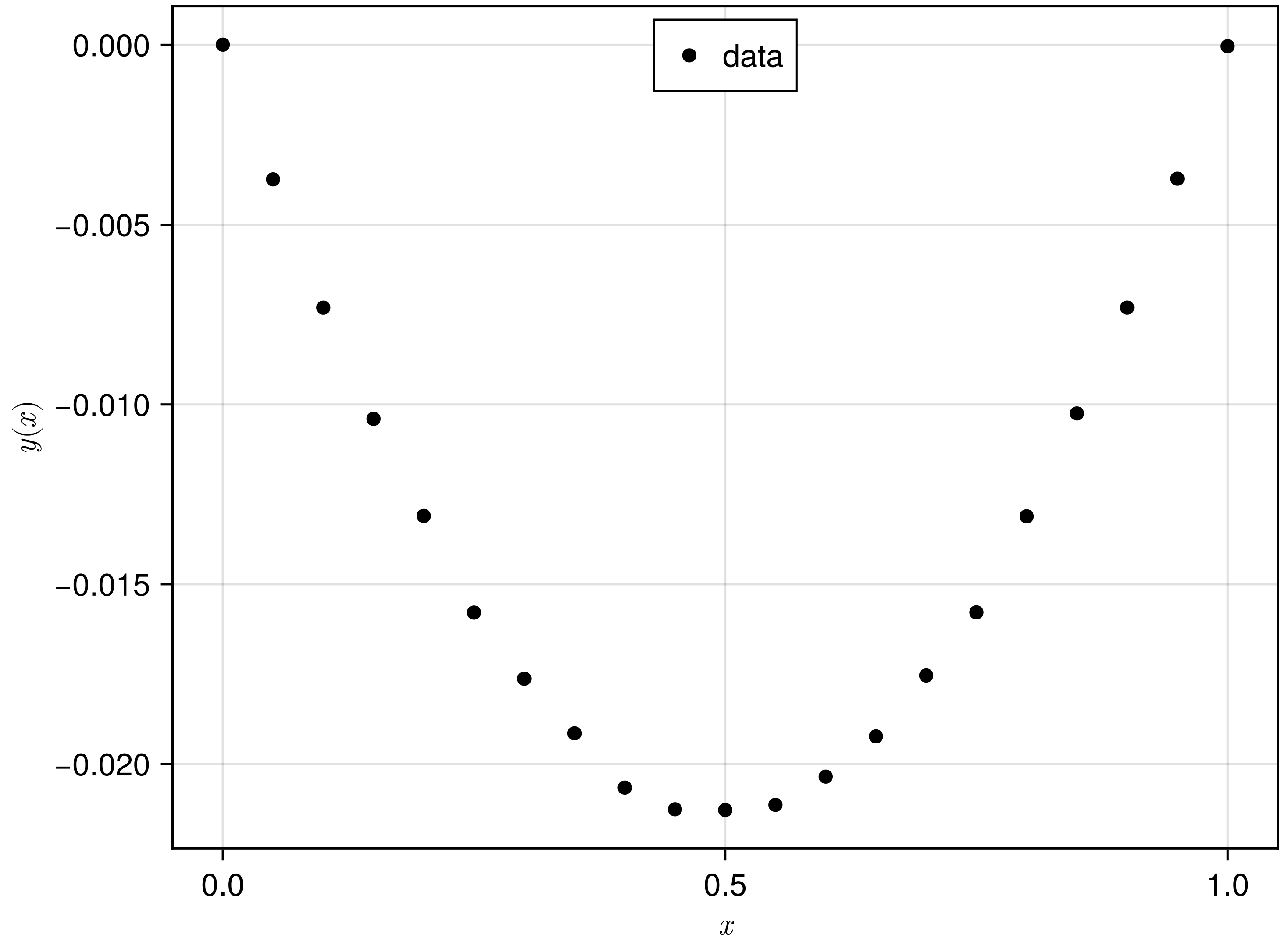
$y(x=0) = 0$



$y(x=1) = 0$

From "science" we know that

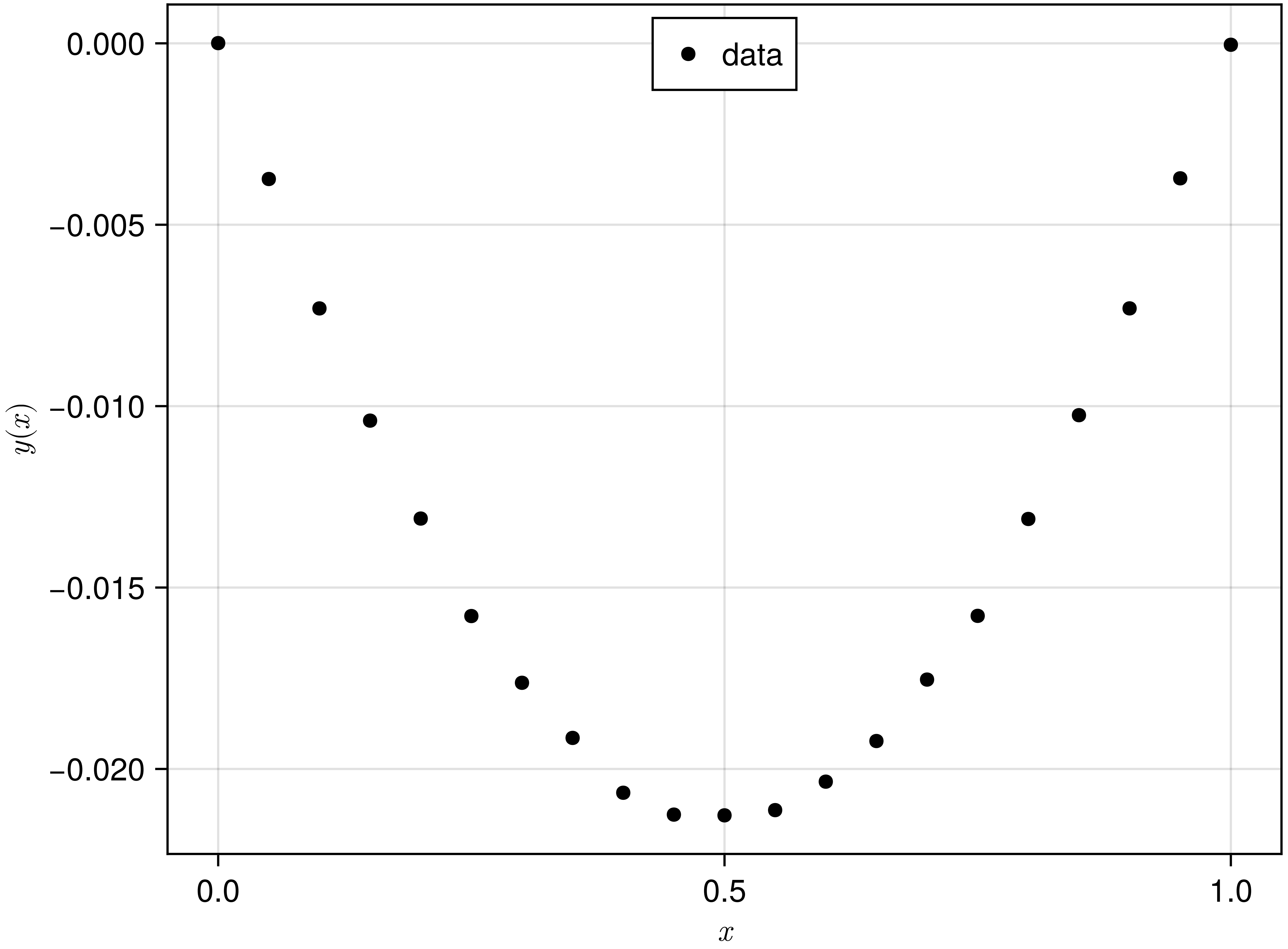
We do not know $w(x)$.....but we have data on y!



$$y(x=0)=0 \qquad \frac{d^2y}{dx^2}+4y(x)=w(x) \qquad y(x=1)=0$$

Use finite difference approximation

$$\frac{y_{i-1}-2y_i+y_{i+1}}{\Delta^2}+4y_i=w(x_i)$$



We do not know $w(x)$ let's just model it as

$$w(x)=a_0+a_1x$$

Thus

$$\frac{y_{i-1}-2y_i+y_{i+1}}{\Delta^2}+4y_i=w(x_i)=a_0+a_1x_i$$

$$\frac{y_{i-1} - 2y_i + y_{i+1}}{\Delta^2} + 4y_i = w(x_i)$$

If we assume we have only 6 points

output input

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1/\Delta^2 & -2/\Delta^2+4 & 1/\Delta^2 & 0 & 0 & 0 \\ 0 & 1/\Delta^2 & -2/\Delta^2+4 & 1/\Delta^2 & 0 & 0 \\ 0 & 0 & 1/\Delta^2 & -2/\Delta^2+4 & 1/\Delta^2 & 0 \\ 0 & 0 & 0 & 1/\Delta^2 & -2/\Delta^2+4 & 1/\Delta^2 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \end{Bmatrix} = \begin{Bmatrix} 0 \\ a_0 + a_1 x_2 \\ a_0 + a_1 x_3 \\ a_0 + a_1 x_4 \\ a_0 + a_1 x_5 \\ 0 \end{Bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1/\Delta^2 & -2/\Delta^2+4 & 1/\Delta^2 & 0 & 0 & 0 \\ 0 & 1/\Delta^2 & -2/\Delta^2+4 & 1/\Delta^2 & 0 & 0 \\ 0 & 0 & 1/\Delta^2 & -2/\Delta^2+4 & 1/\Delta^2 & 0 \\ 0 & 0 & 0 & 1/\Delta^2 & -2/\Delta^2+4 & 1/\Delta^2 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} f_{model}(a_0, a_1, x_1) \\ f_{model}(a_0, a_1, x_2) \\ f_{model}(a_0, a_1, x_3) \\ f_{model}(a_0, a_1, x_4) \\ f_{model}(a_0, a_1, x_5) \\ f_{model}(a_0, a_1, x_6) \end{Bmatrix} = \begin{Bmatrix} 0 \\ a_0 + a_1 x_2 \\ a_0 + a_1 x_3 \\ a_0 + a_1 x_4 \\ a_0 + a_1 x_5 \\ 0 \end{Bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1/\Delta^2 & -2/\Delta^2+4 & 1/\Delta^2 & 0 & 0 & 0 \\ 0 & 1/\Delta^2 & -2/\Delta^2+4 & 1/\Delta^2 & 0 & 0 \\ 0 & 0 & 1/\Delta^2 & -2/\Delta^2+4 & 1/\Delta^2 & 0 \\ 0 & 0 & 0 & 1/\Delta^2 & -2/\Delta^2+4 & 1/\Delta^2 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} f_{model}(a_0, a_1, x_1) \\ f_{model}(a_0, a_1, x_2) \\ f_{model}(a_0, a_1, x_3) \\ f_{model}(a_0, a_1, x_4) \\ f_{model}(a_0, a_1, x_5) \\ f_{model}(a_0, a_1, x_6) \end{Bmatrix} = \begin{Bmatrix} 0 \\ a_0 + a_1 x_2 \\ a_0 + a_1 x_3 \\ a_0 + a_1 x_4 \\ a_0 + a_1 x_5 \\ 0 \end{Bmatrix}$$

So we need to solve to get $f_{model}(a_0, a_1, x_i)$!

Implicit model

$$f_{model}(\vec{a}, x) = a_0$$

$$f_{model}(\vec{a}, x) = a_0 + a_1 x$$

$$f_{model}(\vec{a}, x) = a_0 + a_1 x + a_2 x^2 + \ldots.$$

Explicit Models